

AN AI-BASED WEB APPLICATION FOR SKIN DISEASE PREDICTION & SUGGESTS SOLUTIONS

Project ID: 25-26J-323

Project Proposal Report

Afzal Ahamed AH – IT22102782

B.Sc. (Hons) Degree in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

**AN AI-BASED WEB APPLICATION FOR SKIN DISEASE
PREDICTION & SUGGESTS SOLUTIONS: SKIN DISEASE
PREDICTION**

Project ID: 25-26J-323

Project Proposal Report

Afzal Ahamed AH – IT22102782

B.Sc. (Hons) Degree in Information Technology

Department of Information Technology

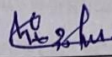
Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

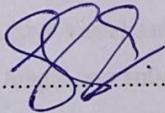
Declaration

I acknowledge that that is my very own work and this proposal does not no longer content without acknowledgment any works formerly submitted for a diploma or degree in every other college or institute of better studying and to the first-rate of our knowledge and notion it does not no longer include any material formerly posted or written through any other man or woman besides in which the acknowledgment is made with inside the text.

Name	Student ID	Signature
Afzal Ahamed. AH	IT22102782	

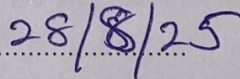
The supervisor/s should certify the proposal report with the following declaration.

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.



Dr. Harinda Fernando

Signature of the supervisor



Date

ACKNOWLEDGEMENT

I sincerely thank everyone who contributed to the completion of my fourth-year research project, *“An AI-based web application for Skin disease prediction & Suggests solutions.”*

My deepest gratitude goes to **Dr. Harindra Fernando**, our project supervisor, for his guidance, support, and insightful feedback throughout this journey. I would also like to thank our team leader, **Vivushan T**, for his leadership, coordination, and encouragement, which were vital in keeping our team focused and productive. I extend my appreciation to the **project panels** for their valuable suggestions, my **project group members** for their collaboration, and **SLIIT** for providing the necessary resources and facilities.

I am also grateful for the opportunity to explore advanced approaches such as **federated learning**, which enhanced the scope and impact of our research. Finally, I sincerely thank my **family and friends** for their constant support and encouragement. This project would not have been possible without all of you.

ABSTRACT

Fungal skin infections represent a pervasive health challenge globally, with heightened prevalence in tropical climates like Sri Lanka, where humidity and warmth accelerate their spread. Conditions such as ringworm (tinea corporis), athlete's foot (tinea pedis), and onychomycosis often mimic other dermatological issues, leading to frequent misdiagnoses and delayed treatments. This proposal outlines the development of an advanced web-based diagnostic system leveraging deep learning (DL) and machine learning (ML) techniques, including convolutional neural networks (CNNs) and support vector machines (SVMs), to enable accurate, real-time prediction of fungal skin diseases. The system incorporates federated learning (FL) for privacy-preserving model training across distributed devices, ensuring data security without centralizing sensitive images. Users can upload skin lesion images and input symptoms in Sinhala, Tamil, or English, receiving personalized predictions, confidence scores, and tailored recommendations such as over-the-counter treatments or lifestyle adjustments. Enhanced preprocessing pipelines, including augmentation and segmentation via U-Net architectures, will improve model robustness against variations in lighting, skin tones, and image quality. By addressing research gaps in privacy, localization for South Asian contexts, and adaptive learning, this application aims to democratize dermatological care, particularly in rural Sri Lanka, reducing misdiagnosis rates and promoting early intervention. The system's backend-frontend synchronization will facilitate intuitive visualizations like heatmaps and dashboards, fostering user trust and educational outreach on fungal prevention.

Index Terms—Skin disease prediction, fungal infections, convolutional neural networks, federated learning, deep learning, privacy-preserving AI, web application.

Table of Contents

ACKNOWLEDGEMENT	3
ABSTRACT	4
LIST OF FIGURES	6
LIST OF TABLES	7
LIST OF ABBREVIATIONS	8
LIST OF APPENDICES	9
1 INTRODUCTION	10
1.1 BACKGROUND.....	10
1.2 LITERATURE SURVEY	11
1.3 RESEARCH GAP	12
1.4 RESEARCH PROBLEM	13
2 OBJECTIVES	14
2.1 MAIN OBJECTIVE	14
2.2 SPECIFIC OBJECTIVES.....	15
3 METHODOLOGIES.....	18
3.1 REQUIREMENTS.....	18
3.1.1 <i>Functional Requirements</i>	18
3.1.2 <i>Non-Functional Requirements</i>	20
3.2 COMMERCIALIZATION	21
3.3 SYSTEM OVERVIEW DIAGRAM	22
4 DECSRIPTION OF PERSONAL AND FACILITIES	22
5 REFERENCES	25
APPENDICES	27
CONCLUSION.....	28

LIST OF FIGURES

Page

Figure 3.3.1	System high level diagram.....	22
Figure 3.3.3	High level diagram of individual function.....	20

LIST OF TABLES

Table 1.3.1	Comparison of existing systems	12
Table 4.1	Description of personal and facilities	24
Table 5.1	Budget per month	24

LIST OF ABBREVIATIONS

Abbreviation	Description
DL	Deep Learning
ML	Machine Learning
CNN	Convolutional Neural Networks
SVM	Support Vector machines
AI	Artificial Intelligence
FL	Federated Learning

LIST OF APPENDICES

Appendix	Description	Page
Appendix – A	Work breakdown chart	23
Appendix – B	Gantt chart	23

1 INTRODUCTION

1.1 Background

Fungal skin diseases pose a substantial burden on global public health, affecting an estimated billion annually and manifesting in forms ranging from superficial irritations to invasive infections that can compromise vital organs [1]. In tropical environments such as Sri Lanka, environmental factors like high humidity and frequent monsoons exacerbate the incidence of pathogens including dermatophytes and *Candida* species, leading to conditions like tinea corporis (ringworm), tinea pedis (athlete's foot), tinea cruris (jock itch), and onychomycosis (nail fungus) [2]. These infections not only cause physical symptoms such as itching, scaling, and discoloration but also contribute to psychological distress, social stigma, and economic strain due to recurrent treatments and lost productivity. Vulnerable populations, including immunocompromised individuals, agricultural workers, and those in overcrowded urban settings, face amplified risks, with misdiagnosis rates reaching up to 30% in primary care settings owing to visual similarities with eczema, psoriasis, or bacterial infections [3].

Traditional diagnostic methods rely heavily on clinical expertise, microscopic examinations, or cultures, which are resource-intensive and often inaccessible in rural Sri Lanka, where dermatologist-to-patient ratios remain alarmingly low [4]. This disparity underscores the need for innovative, technology-driven solutions that can bridge healthcare gaps. Artificial intelligence (AI), particularly DL and ML, has emerged as a transformative force in medical imaging, demonstrating capabilities to rival or surpass human accuracy in classifying dermatological conditions [5]. Building on this, the proposed web application integrates advanced AI models to predict fungal skin diseases from user-uploaded images and symptom descriptions, while suggesting evidence-based solutions. By incorporating FL, the system ensures privacy by training models locally on user devices without transmitting raw data, addressing ethical concerns in health AI [6]. This approach not only enhances accessibility but also adapts to diverse Sri Lankan demographics, including varying skin tones (Fitzpatrick types IV-VI prevalent in South Asia) and multilingual support for Sinhala, Tamil, and English inputs.

The application's core innovation lies in its adaptive DL framework, which uses federated updates to refine models across clients, recognizing subtle fungal patterns in diverse environments such as humid rural areas or urban pollution-exposed skin. This proposal expands on prior discussions by integrating hybrid CNN-ML architectures for robust detection, advanced preprocessing for image quality enhancement, and interactive visualizations for user engagement. Ultimately, this system aims to empower users with timely insights, reduce healthcare disparities, and contribute to public health initiatives in Sri Lanka.

1.2 Literature Survey

Extensive research has demonstrated the efficacy of AI in dermatological diagnostics, with a focus on image-based classification using DL techniques. For instance, Esteva . [7] developed a CNN-based system for skin cancer detection, achieving dermatologist-level accuracy on dermoscopic images, highlighting the potential of transfer learning from large datasets like ImageNet. This work inspired adaptations for non-melanoma conditions, including fungal infections, where visual cues like annular lesions or hyphae structures are key discriminants.

In the context of infectious diseases, Ozturk . [8] applied CNNs to chest X-rays for COVID-19 detection, underscoring the adaptability of DL to radiological patterns, a principle extendable to dermatology. Muhaba . [9] proposed a mobileNet-v2 model for classifying five common skin diseases using clinical images and patient metadata, attaining 97.5% accuracy through preprocessing techniques like resizing and augmentation. Their emphasis on smartphone integration aligns with our web application's accessibility goals, though it lacks privacy mechanisms like FL.

Further advancements include hybrid approaches, as seen in Alenezi . [10], who combined CNNs with ANNs for melanoma recognition on PH2 and ISIC datasets, yielding 98% accuracy. This validates the use of ensemble methods for fungal detection, where subtle features require multi-architecture fusion. Sri Lankan-specific studies, such as Thapalingasivam . [11], utilized ensemble DL (VGG16, EfficientNetB0) on local datasets from Vavuniya Hospital, achieving 91% accuracy for conditions like tinea, tailored to South Asian skin types. Wijesinghe . [12] reviewed image processing for skin disease identification in Sri Lanka, noting high prevalence (47.6% in semi-urban areas) and advocating ML integration.

Federated learning has addressed privacy gaps in medical AI. Sharif . [13] surveyed FL for medical imaging, emphasizing decentralized training to comply with regulations like GDPR. In dermatology, Zhang [14] applied FL to skin cancer diagnosis, improving generalization without data sharing. These insights fill research gaps in our proposal, where FL enables secure, adaptive training for fungal patterns across diverse Sri Lankan devices.

Despite these advances, limitations persist: most systems overlook fungal-specific subtlety, multilingual inputs, and real-time recommendations. Our enhanced system bridges this by incorporating U-Net segmentation for lesion isolation, GAN-based augmentation for data diversity, and backend-frontend sync for visualizations like Grad-CAM heatmaps, ensuring clinical relevance and user

1.3 Research gap

Based on the literature review, several research gaps exist in using web applications to predict fungal skin diseases through machine learning and deep learning, particularly regarding data protection and patient privacy. Fungal diseases are a major public health concern, affecting millions worldwide, ranging from mild skin infections to severe, life-threatening systemic conditions such as candidemia, aspergillosis, and cryptococcosis. Immunocompromised individuals, including cancer patients and organ transplant recipients, are especially vulnerable. The increasing incidence of fungal infections is driven by factors such as immunosuppressive therapies, aging populations, misuse of antifungal drugs, and environmental changes like climate change and urbanization.

Early detection, accurate diagnosis, and timely treatment are essential, and advances in molecular diagnostics, imaging, and computational approaches are improving pathogen identification. Preventive strategies, patient awareness, and appropriate therapies remain crucial in reducing the health burden.

This portal provides a comprehensive resource on human fungal diseases, focusing on identification, prevention, treatment, and overall public health impact, leveraging modern technology to support both healthcare providers and patients.

	Research 1	Research 2	Research 3	Research 4	Our Proposed System
Skin Disease Prediction using Image pre- processing	✗	✗	✓	✓	✓
Deep Learning Based approach	✗	✓	✓	✗	✓
Federated Learning Based approach	✗	✗	✗	✗	✓
Web application	✓	✓	✗	✓	✓
User-uploaded photos, and clinical data	✗	✗	✗	✓	✓

Figure 1.3.1: Comparison of existing systems

1.3 Research Problem

From the literature, several gaps emerge in AI-driven fungal skin disease diagnostics. Existing systems largely focus on general dermatology or skin cancer, underrepresenting fungal infections despite their significant global burden [15]. Privacy concerns are also inadequately addressed, as many solutions rely on centralized data storage, creating risks of sensitive health information breaches [16]. Furthermore, most models lack localization for South Asian populations, including Sri Lanka, where fungal skin diseases are highly prevalent due to tropical environmental factors. Current systems rarely incorporate regional datasets or multilingual symptom processing, leading to diagnostic biases, especially for individuals with darker skin tones [17].

The research problem is underscored by high misdiagnosis rates (20–30%) for fungal infections in Sri Lanka, a situation exacerbated by limited access to dermatologists, particularly in rural areas [18]. Existing diagnostic tools depend heavily on manual expertise, resulting in delayed interventions, increased costs, and recurring infections. While some AI-based systems have been proposed, their accuracy remains limited, and they fail to offer real-time, accessible solutions suitable for resource-constrained environments.

Another pressing challenge is the absence of privacy-preserving mechanisms and real-time diagnostic capabilities in current models. Early detection is crucial for fungal skin diseases since delayed treatment often leads to complications, higher healthcare expenses, and greater psychological and social impacts. However, most current systems are unable to deliver instant, user-friendly predictions, reducing their practical applicability.

Given these challenges, there is a clear need for an accurate, privacy-preserving, real-time web application that predicts fungal skin diseases, provides tailored treatment suggestions, and adapts to local environments through federated learning. Such a system would not only address diagnostic gaps and reduce dependence on specialist expertise but also ensure equitable healthcare access across Sri Lanka. Implementing this solution could significantly improve patient outcomes, reduce the national healthcare burden, and mitigate the social stigma associated with fungal skin diseases.

2 OBJECTIVES

2.1 Main objective

The main objective of this study is to develop an AI-based web application for the accurate prediction and management of fungal skin diseases using machine learning, specifically Convolutional Neural Networks (CNNs). The system will enable users to upload skin images and provide symptom descriptions in Sinhala, Tamil, or English, offering real-time diagnosis of fungal infections such as ringworm (*tinea corporis*), athlete's foot (*tinea pedis*), and nail fungus (*onychomycosis*). By incorporating image preprocessing and data augmentation to handle diverse skin types and conditions, the system aims to achieve high diagnostic accuracy. It will provide immediate feedback with confidence scores and personalized recommendations, such as antifungal treatments and hygiene tips, to support early intervention. Designed to improve access to dermatological care in Sri Lanka, particularly in rural areas, this application seeks to reduce misdiagnosis, enhance patient outcomes, and promote awareness of fungal skin conditions, ultimately improving quality of life for users.

2.2 Specific objectives

2.2.1 Create a detailed image dataset.

To ensure that the sample is robust and inclusive, collect skin images representing a wide range of skin conditions, including skin types, age groups and gender.

The process of collecting a dataset for predicting fungal skin disease.

Definition of Scope: Specify a general list, including the types of skin diseases that need to be covered, skin conditions, and any specific characteristics or demographics (e.g., age, gender, race).

Source Information: Identify and collect data sources from which can compile a picture of skin diseases. These resources can include medical institutions, dermatology clinics, publicly available datasets, or even user-generated content from mobile apps (with appropriate consent and privacy considerations).

Data selection: Carefully select and maintain images that represent a wide range of skin tones. Ensure that the dataset includes skin types, age groups, genders, and ethnic groups so that it is comprehensive and unbiased.

Pre-processing data: Match images, sizes, and styles to ensure consistency. Typical pre-processing steps include amplification, normalization, and noise reduction.

Description and Labelling: Each image should be carefully annotated and labelled with the corresponding skin condition. Accurate scripts are important for supervised learning. This can be done manually by dermatologists or crowdsourcing, but it takes expertise to ensure it is correct.

Dataset classification: Divide the dataset into two parts: training and testing. A training set is used for model training and a test set for model performance evaluation

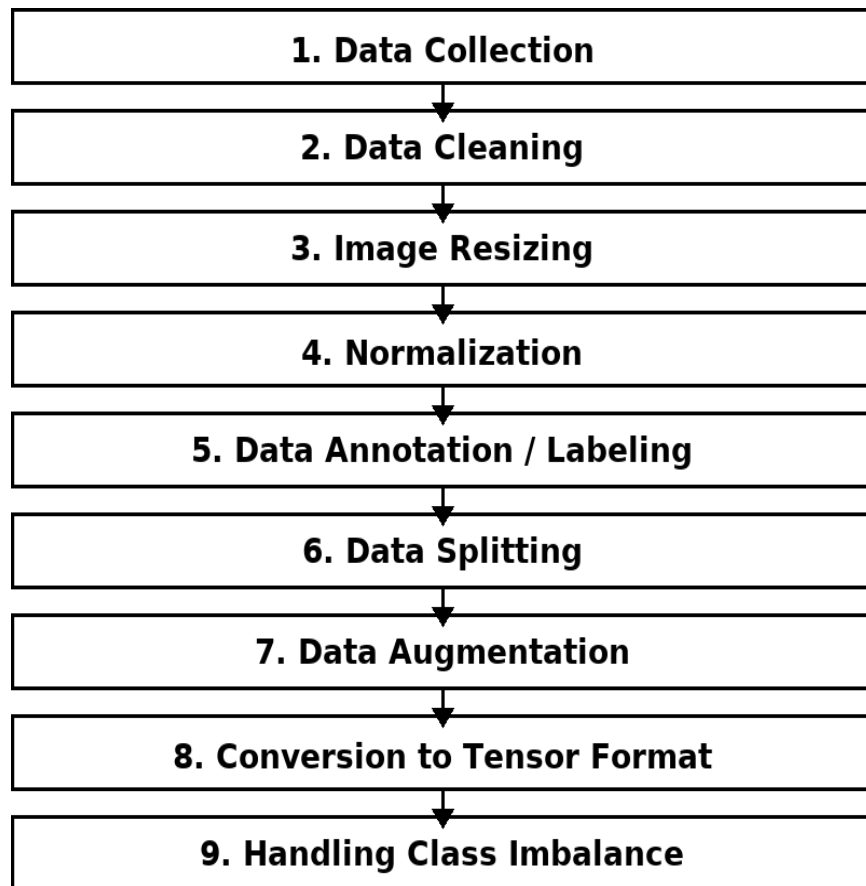


Figure 2.2.1.1: Data pre-processing flow

2.2.2 Data Pre-processing and Augmentation

Implement data pre-processing techniques to standardize image sizes, enhance image quality, and prepare the dataset for CNN training. Apply data augmentation methods to expand the dataset, improve model generalization, and account for variations in image capture conditions.

2.2.3 Training a CNN Model

Design and train a CNN-based machine learning model using the prepared image dataset to classify fungal skin diseases accurately. Fine-tune the model for optimal performance, accuracy, and efficiency.

2.2.4 Enable Real-Time Prediction

Integrate the trained CNN model into the system to enable real-time predictions. Ensure that the system can process user-uploaded images and symptoms swiftly and provide immediate feedback.

2.2.5 Accuracy and Evaluation

Conduct rigorous testing and evaluation to assess the accuracy and reliability of the skin disease prediction system. Explore opportunities for validation in collaboration with dermatologists or medical experts to validate the system's clinical relevance.

3 METHODOLOGIES

3.1 Requirements

3.1.1 Functional Requirements

Image Upload and Description:

- Users should be able to upload skin images.
- Users should provide symptom descriptions in English.

Real-Time Prediction:

- Enable real-time predictions for users.
- The system should accept user-uploaded images and symptom descriptions and provide immediate feedback.

Prediction Accuracy:

- The system should achieve a specified level of accuracy in predicting skin diseases based on medical standards.

Data Collection and Management:

- The system should collect and manage a diverse dataset of skin images representing various skin conditions, types, ages, genders, and ethnicities.
- It should allow for the addition of new data over time.

Data Pre-processing and Augmentation:

- Automatically standardize image sizes and enhance quality during data pre-processing.
- Apply data augmentation techniques to expand the dataset.

CNN Model Training:

- Train a Convolutional Neural Network (CNN) model using the prepared dataset.
- Fine-tune the model to achieve optimal accuracy and efficiency.

Federated Learning:

- Use federated learning for collaborative model training across devices/institutions, enhancing generalization while preserving data privacy

Disease Solutions and Recommendations:

- Provide general treatment info, home remedies, and preventive tips based on medical guidelines, with disclaimers urging consultation with dermatologists.

Data Privacy:

- User data, including images and symptoms, should be handled securely and in compliance with privacy regulations.

Educational Content:

- Provide educational materials within the system to inform users about the prediction process, its limitations, and the importance of consulting medical professionals.

Continuous Improvement:

- The system should be designed to allow for continuous model improvement based on user feedback and emerging medical knowledge.

3.1.2 Non-Functional Requirements

Performance:

- The system should respond to user queries and image uploads swiftly, aiming for real-time or near-real-time processing.
- It should handle a substantial user load without significant performance degradation.

Accuracy:

- The CNN model should achieve a high level of accuracy in classifying skin diseases.
- The accuracy should be regularly monitored and improved.

Scalability:

- The system should be designed to scale horizontally to accommodate growing data and user loads.

Security:

- Ensure data security and privacy compliance, especially for user-uploaded images and personal information.
- Implement robust authentication and authorization mechanisms.

Reliability:

- The system should be available 24/7 with minimal downtime for maintenance.
- Implement backup and disaster recovery procedures.

Usability:

- The user interface should be intuitive and user-friendly.
- Provide clear instructions for image upload and symptom description.

3.2 Commercialization

The proposed Skin Fungal Disease Detection System utilizes Convolutional Neural Networks (CNNs) and advanced image processing techniques to accurately identify various skin fungal infections. This innovation addresses the growing demand for accessible, reliable, and real-time dermatological care, especially in remote and underserved regions where specialized healthcare is limited.

The system's commercialization strategy should include partnerships with healthcare institutions, telemedicine platforms, and mobile health apps to reach a wide user base. Revenue streams can be generated through subscription models, licensing agreements with healthcare providers, or even offering premium services for in-depth skin analysis.

Moreover, collaborations with dermatologists and medical experts will lend credibility to the system, making it an invaluable tool in clinical practice. Ensuring compliance with healthcare data regulations will build trust among users and stakeholders.

The commercialization of this technology not only promises financial success but also contributes significantly to improving overall healthcare outcomes and the quality of life for users. It empowers individuals to take control of their skin health, enabling early intervention and potentially revolutionizing the field of skin healthcare.

3.3 system overview diagram

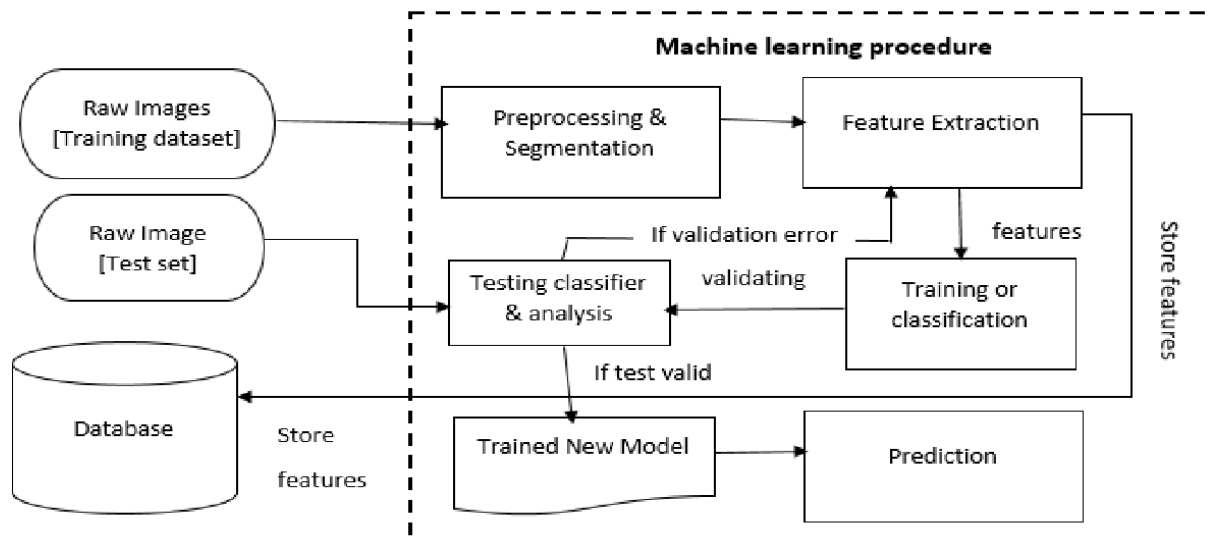


Figure 3.3.1: System Architecture

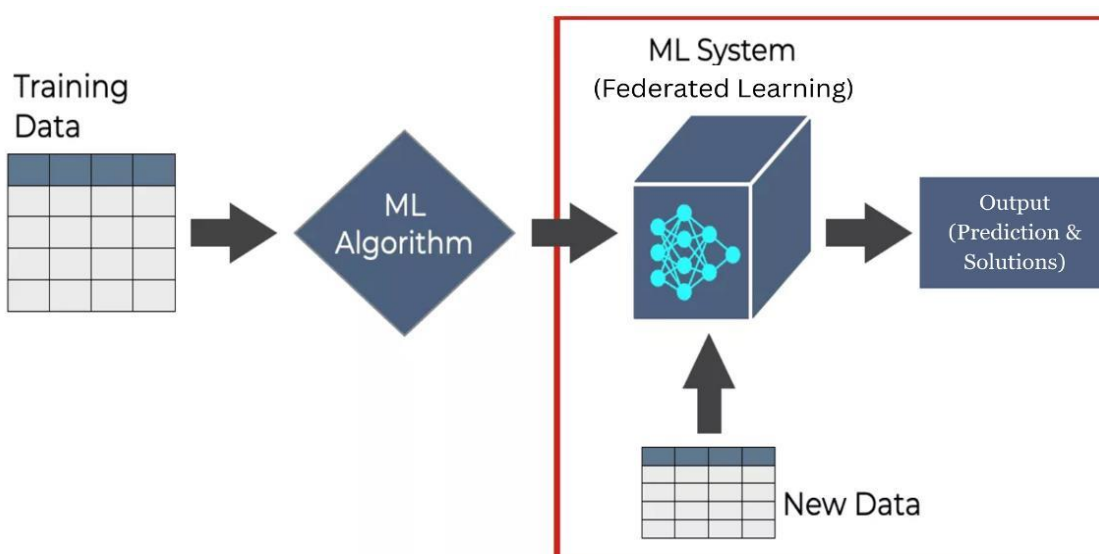


Figure 3.3.2: Prediction System

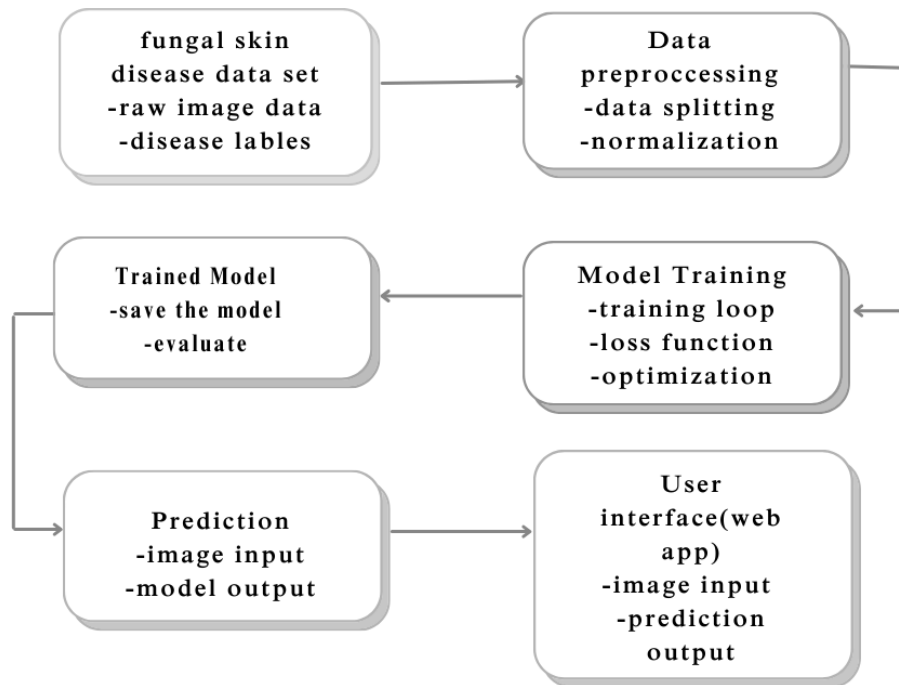


Figure 3.3.3: Individual component diagram



Figure 3.3.4: Individual workflow diagram

4 DECSRIPTION OF PERSONAL AND FACILITIES

Registration Number	Name	Functions / Task
IT22102782	Afzal Ahamed AH	• Dataset Creation or Collection • Processing the data • Apply CNN algorithm. • Trained the model. • Implement the model for fungal skin disease prediction • Suggests solutions

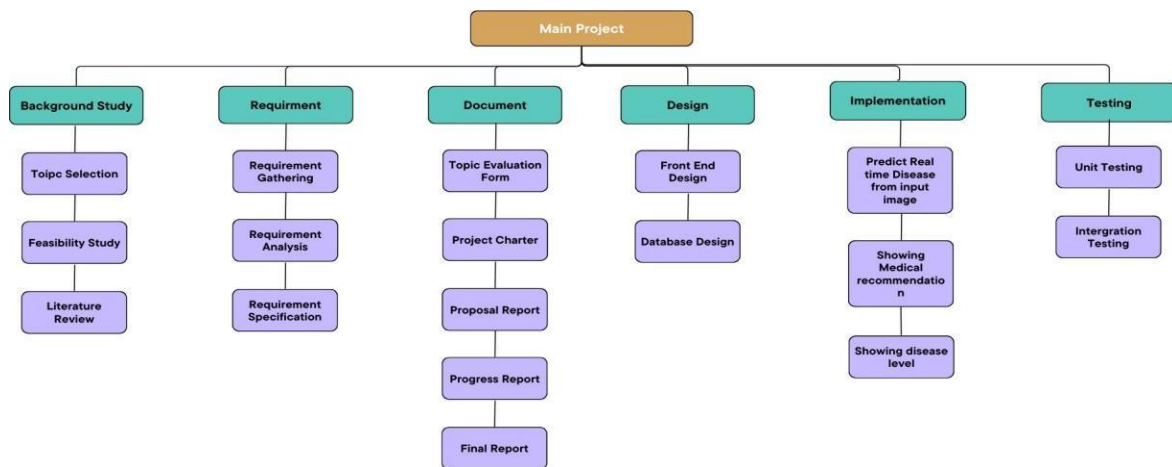
Table 4.1: Description of personal and facilities

5 REFERENCES

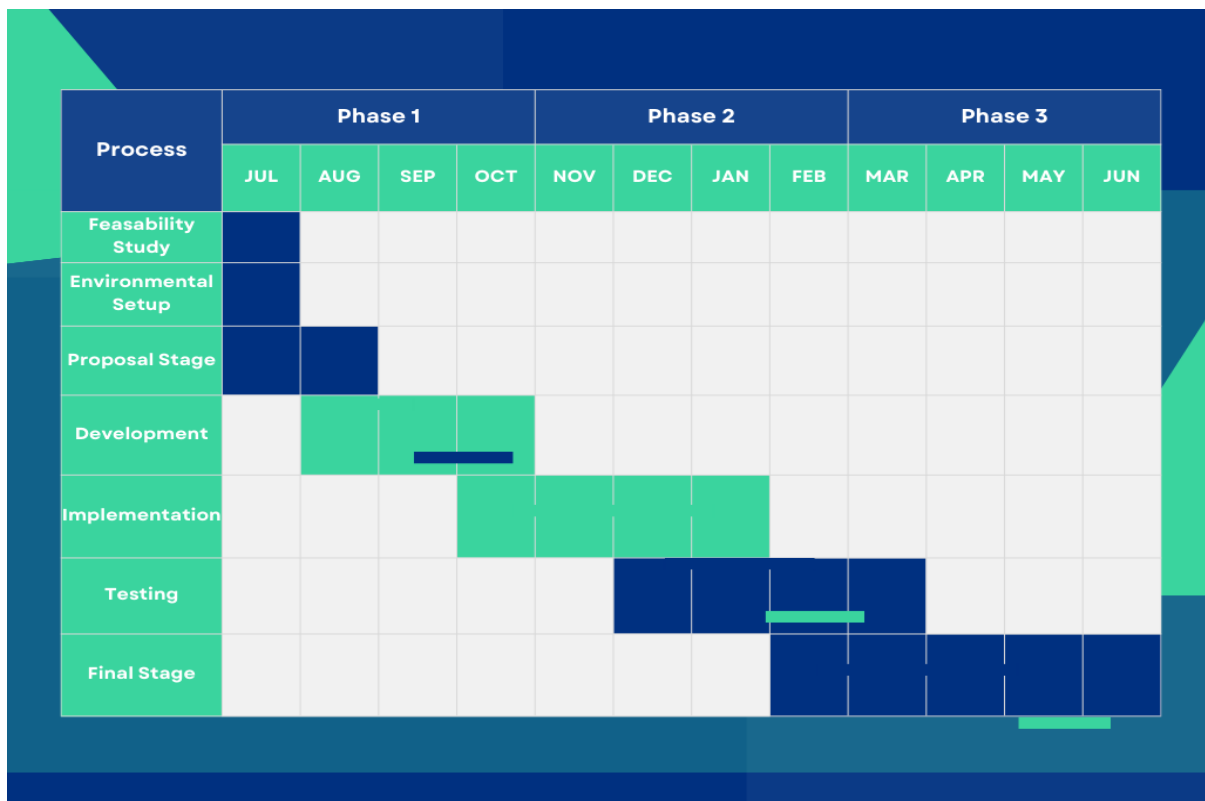
- [1] P. Chanyachailert, C. Leeyaphan, and S. Bunyaratavej, "Cutaneous fungal infections caused by dermatophytes and non-dermatophytes: An updated comprehensive review of epidemiology, clinical presentations, and diagnostic testing," *J. Fungi*, vol. 9, no. 6, p. 669, Jun. 2023, doi: 10.3390/jof9060669.
- [2] D. A. Q. Shakir and E. T. Mahdi, "Machine learning techniques for skin fungal infection detection—A review," *Mesopotamian J. Artif. Intell. Healthcare*, vol. 2024, pp. 177–183, Dec. 2024, doi: 10.58496/MJAIH/2024/018.
- [3] T. Debebe and B. Molla, "Human skin fungal diseases classification using deep learning technique," *Harla J. Eng. Technol.*, vol. 1, no. 2, pp. 16–37, 2022.
- [4] K. Thapalingasivam, V. Vinoharan, and P. Sivanesan, "Classification of Sri Lankan skin diseases using ensemble deep learning models," in *Proc. Int. Conf. Inf. Technol. Res. (ICITR)*, Univ. Vavuniya, Sri Lanka, 2024, pp. 1–6. [Online]. Available: http://drr.vau.ac.lk/bitstream/handle/123456789/1161/2024_Kayalvizhi_ICITR2024_Publication_compressed%20-%20Veerapathirapillai%20Vinoharan.pdf
- [5] L. A. N. Wijesinghe, D. M. R. Kulasekera, and W. M. K. S. Ilmini, "Research directions for skin disease identification using image processing and machine learning," in *Int. Conf. Adv. Res. Comput. (ICARC)*, Gen. Sir John Kotelawala Defence Univ., Sri Lanka, 2021, pp. 1–6. [Online]. Available: <http://ir.kdu.ac.lk/bitstream/handle/345/2283/com032.pdf>
- [6] M. I. Sharif, M. Mehmood, M. P. Uddin, K. Siddique, Z. Akhtar, and S. Waheed, "Federated learning for analysis of medical images: A survey," *J. Comput. Sci.*, vol. 20, no. 12, pp. 1610–1621, Oct. 2024, doi: 10.3844/jcssp.2024.1610.1621.
- [7] A. Esteva., "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, Feb. 2017, doi: 10.1038/nature21056.
- [8] T. Ozturk, M. Talo, E. A. Yildirim, U. B. Baloglu, O. Yildirim, and U. R. Acharya, "Automated detection of COVID-19 cases using deep neural networks with X-ray images," *Comput. Biol. Med.*, vol. 121, p. 103792, Jun. 2020, doi: 10.1016/j.compbimed.2020.103792.
- [9] K. A. Muhaba, K. Dese, T. M. Aga, F. T. Zewdu, and G. L. Simegn, "Automatic skin disease diagnosis using deep learning from clinical image and patient information," *Skin Res. Technol.*, vol. 28, no. 3, pp. 493–503, May 2022, doi: 10.1111/srt.13148.
- [10] F. Alenezi, "Developing a recognition system for diagnosing melanoma skin lesions using artificial intelligence algorithms," *Comput. Math. Methods Med.*, vol. 2021, p. 9998379, May 2021, doi: 10.1155/2021/9998379.
- [11] K. Thapalingasivam, V. Vinoharan, and P. Sivanesan, "Classification of Sri Lankan skin diseases using ensemble deep learning models," in *Proc. Int. Conf. Inf. Technol. Res. (ICITR)*, Univ. Vavuniya, Sri Lanka, 2024, pp. 1–6.
- [12] L. A. N. Wijesinghe, D. M. R. Kulasekera, and W. M. K. S. Ilmini, "Research directions for skin disease identification using image processing and machine learning," in *Int. Conf. Adv. Res. Comput. (ICARC)*, Gen. Sir John Kotelawala Defence Univ., Sri Lanka, 2021, pp. 1–6.
- [13] M. I. Sharif., "Federated learning for analysis of medical images: A survey," *J. Comput. Sci.*, vol. 20, no. 12, pp. 1610–1621, Oct. 2024.
- [14] Y. Zhang, "Distributed learning in healthcare: Application of federated learning to skin cancer diagnosis," in *Proc. 1st Int. Conf. Eng. Manage., Inf. Technol. Intell. (EMITI 2024)*, 2024, pp. 675–678, doi: 10.5220/0012967700004508.
- [15] T. D. Nigat, T. M. Sitote, and B. M. Gedefaw, "Fungal skin disease classification using the convolutional neural network," *J. Healthcare Eng.*, vol. 2023, p. 6370416, May 2023, doi: 10.1155/2023/6370416.

- [16] L. Ju ., "Federated learning for medical image analysis: A survey," *Artif. Intell. Med.*, vol. 148, p. 102769, Feb. 2024, doi: 10.1016/j.artmed.2024.102769.
- [17] M. W. P. Maduranga and D. Nandasena, "Mobile-based skin disease diagnosis system using convolutional neural networks (CNN)," *Int. J. Image, Graph. Signal Process. (IJIGSP)*, vol. 14, no. 3, pp. 47–57, Jun. 2022, doi: 10.5815/ijigsp.2022.03.05.
- [18] A. De, A. Sarda, S. Gupta, and S. Das, "Use of artificial intelligence in dermatology," *Indian Dermatol. Online J.*, vol. 11, no. 5, pp. 835–840, Sep. 2020, doi: 10.4103/idoj.IDOJ_692_19.

APPENDICES



Appendix – A: Work breakdown chart



Appendix – B: Gantt chart

Conclusion

Fungal infections of the skin represent a significant public health concern worldwide, particularly in tropical regions such as Sri Lanka, where hot and humid conditions create an environment conducive to fungal growth. Untreated or misdiagnosed fungal skin conditions can lead to severe complications, discomfort, and social stigma, impacting the overall well-being and quality of life of affected individuals.

Recent advancements in machine learning and computer vision provide promising solutions for improving dermatological healthcare through early detection and accurate diagnosis of fungal infections. This research aims to address Sri Lanka's specific healthcare challenges by developing a web-based application that utilizes deep learning models to predict and classify common human skin fungal diseases, while incorporating local language support and health awareness programs relevant to the region.

Through this initiative, we seek to enhance accessibility to dermatological services, raise awareness about fungal skin infections, and enable early intervention to prevent severe complications. We strongly believe that technology can play a critical role in bridging the gap between patients and dermatological expertise, particularly in underserved and rural areas.

Although progress has been made in applying machine learning to dermatological conditions, research gaps remain in areas such as data privacy and security, interpretability of deep learning models, clinical validation, real-time telemedicine integration, and ethical guidelines for AI use in healthcare. Addressing these challenges will be essential for ensuring responsible and trustworthy deployment of technology in dermatology, ultimately improving healthcare outcomes and empowering individuals to manage their skin health effectively.

About this page

This is your assignment dashboard. You can upload submissions for your assignment from here. When a submission has been processed you will be able to download a digital receipt, view any grades and similarity reports that have been made available by your instructor.

> Research Paper Checking ?

Paper Title	Uploaded	Grade	Similarity
IT22102782_Afzal.pdf	08/27/2025 8:09 PM	--	13%   